Experiment No. 22

Write the Prolog program to implement family tree.

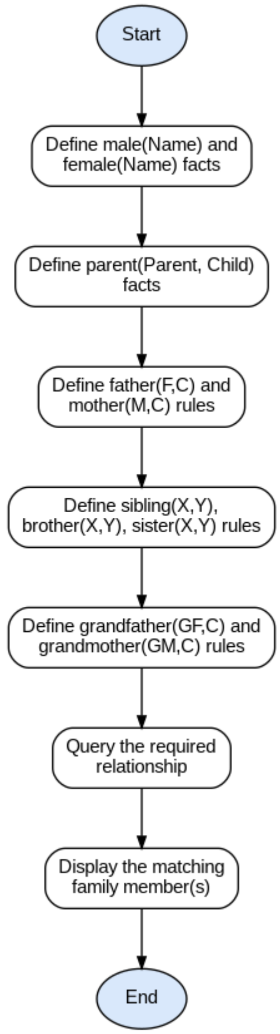
Aim

To write a Prolog program that represents a family tree using facts, and to define rules for relationships such as father, mother, brother, sister, grandfather and grandmother.

Algorithm

1. Start.
2. Define facts male(Name) and female(Name) to record the gender of each family member.
3. Define facts parent(Parent, Child) to record the parent-child relationships in the family.
4. Define the rule father(F, C): F is a parent of C and F is male.
5. Define the rule mother(M, C): M is a parent of C and M is female.
6. Define the rule sibling(X, Y): X and Y share a common parent and X is not the same as Y.
7. Define the rule brother(X, Y): X is a sibling of Y and X is male.
8. Define the rule sister(X, Y): X is a sibling of Y and X is female.
9. Define the rule grandfather(GF, C): GF is a parent of some P, and P is a parent of C, and GF is male.
10. Define the rule grandmother(GM, C) in the same way, using female instead of male.
11. Query each relationship for the desired family members and display the results.
12. Stop.

Flowchart



Program

```
male(john).
male(steve).
male(alex).
female(mary).
female(linda).
female(anna).

parent(john, steve).
parent(john, linda).
parent(mary, steve).
parent(mary, linda).
parent(steve, alex).
parent(anna, alex).
```

```
father(F, C) :-
   parent(F, C),
   male(F).

mother(M, C) :-
   parent(M, C),
   female(M).

sibling(X, Y) :-
   parent(P, X),
   parent(P, Y),
   X \= Y.

brother(X, Y) :-
   sibling(X, Y),
   male(X).

sister(X, Y) :-
   sibling(X, Y),
   female(X).

grandfather(GF, C) :-
   parent(GF, P),
   parent(P, C),
   male(GF).

grandmother(GM, C) :-
   parent(GM, P),
   parent(P, C),
   female(GM).

main :-
   father(F, alex), format("~w is the father of alex~n", [F]),
   mother(M, steve), format("~w is the mother of steve~n", [M]),
   sister(linda, steve), format("linda is the sister of steve~n"),
   brother(steve, linda), format("steve is the brother of linda~n"),
   grandfather(GF, alex), format("~w is the grandfather of alex~n", [GF]),
   grandmother(GM, alex), format("~w is the grandmother of alex~n", [GM]).
```

Result

The program was executed successfully and the output was verified as correct.

?- main.
steve is the father of alex
mary is the mother of steve
linda is the sister of steve
steve is the brother of linda
john is the grandfather of alex
mary is the grandmother of alex
true.

Sample Output — Running in VS Code

Screenshot of the program being run in the Visual Studio Code integrated terminal: